Experiment 1: ML Project Structure with Git, Conda & Azure ML Integration

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AIM

To study the theoretical setup of a reproducible Machine Learning (ML) project using Git and Conda and its integration with Azure Machine Learning Workspace.

OBJECTIVES

1. To understand the importance of reproducibility in ML projects.
2. To study the role of Git and Conda in managing code and dependencies.
3. To understand how Azure ML Workspace and Azure ML Environments support cloud-based ML workflows.

THEORY

1. Reproducibility and Project Structure

Reproducibility means obtaining the same results using the same data, code, and environment. In ML, results may change due to different software versions or system settings. Therefore, proper management of code and dependencies is essential.

A structured ML project separates data, source code, models, and documentation. This improves clarity, collaboration, and maintainability while ensuring reliable and consistent results.

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2. Role of Git and Conda (Offline Setup)

Git is a version control system that tracks changes in project files. It helps manage different experiment versions, maintain history, and restore previous stable states.

Conda is used to create isolated environments with specific Python and library versions. This prevents dependency conflicts and ensures the same environment can be recreated on another system.

Jupyter Notebook and Google Colab are commonly used for experimentation and testing during offline development.

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3. Azure ML Workspace

Microsoft Azure provides

Azure Machine Learning for building and managing ML models in the cloud.

An Azure ML Workspace acts as a centralized platform to manage datasets, experiments, models, and compute resources. It supports scalable training, collaboration, and experiment tracking.

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4. Azure ML Environments and Workflow

Azure ML Environments define the runtime configuration for training and deployment, including Python version and required libraries. This ensures consistent and reproducible cloud execution.

In practice, development is done offline using Git and Conda, and then integrated with Azure ML for large-scale training and deployment. This combination provides flexibility, scalability, and a structured ML lifecycle.

CONCLUSION

In this experiment, we studied the theoretical concepts of setting up a reproducible ML project using Git and Conda and integrating it with Azure Machine Learning Workspace.

Combining offline development tools with Azure ML ensures a structured, scalable, and reproducible Machine Learning workflow suitable for real-world applications.